

DREAMER CUBE

PHASE 1 BUILD GUIDE

One battery-powered, solar-ready autonomous fruiting chamber

Prototype objective: prove the climate system before building the finished cabinet

DESIGN BASIS

CHAMBER	POWER	CHARGING	CONTROL
80-qt gasket tote	Bluetti AC70	200W solar	Inkbird + sensors

Phase 1 finished result

- One 80-quart gasketed tub in front-opening sideways orientation
- One removable grow tray or cassette with a raised internal shelf
- One low-speed circulation fan plus passive fresh-air openings
- External ultrasonic humidifier controlled by an Inkbird IHC-200
- Bluetti AC70 battery power with solar charging capability
- Basic temperature, humidity and energy logging
- Measured battery runtime and daily energy use before scaling

Success criteria

☐	Stable humidity without direct fog striking the substrate
☐	Gentle air movement without excessive surface drying
☐	Controlled condensate and dry electrical connections
☐	Overnight battery coverage and measurable solar recovery
☐	Automatic restart after power interruption
☐	At least 24 hours of safe unattended operation
☐	Easy cleaning, tray removal and component access

1. Choose the test location

Use a garage, basement, utility room or sheltered outdoor work area for initial testing.

- Allow roughly 36 in wide x 30 in deep x 48 in high.
- Provide access to sunlight for the solar panel.
- Keep a dry, shaded location available for the AC70.
- Avoid direct vehicle exhaust and rain exposure.
- Keep the battery outside the humidified chamber.

2. Buy the Phase 1 hardware

Purchase only the components needed to prove one complete chamber.

Subsystem	Required components
Chamber	80-qt gasket-lid tote; wire shelf; shallow grow tray; drip tray; stainless hardware; neutral-cure silicone; PTFE tape
Airflow	Adjustable low-speed fan; fan guard; coarse washable filter; filter patches or micropore tape
Humidity	Ultrasonic humidifier with auto-restart; Inkbird IHC-200; silicone hose; bulkhead/barb; clamps
Power	Bluetti AC70; 200W panel; compatible solar cable; grounded power strip; plug-in watt meter
Monitoring	Independent digital temp/RH sensor, or ESP32 plus quality temp/RH sensor and USB cable

3. Bench-test every electrical component

Test the equipment before drilling or modifying the tub.

- Fully charge the AC70.
- Connect and test the fan at its lowest setting.
- Plug the humidifier into the Inkbird humidity-controlled outlet.
- Confirm the humidifier resumes automatically after power is restored.
- Measure each component individually with the watt meter.

BUILD NOTE: Do not drill the tub until the humidifier passes the automatic-restart test.

4. Rotate and support the tub

Set the tote on its side so the gasket lid opens toward you.

- Support it on a flat shelf or two rails.
- Confirm the lid opens fully and the gasket remains seated.
- Check the shell for bowing before adding load.
- Keep all mounting temporary until fit is confirmed.

5. Install the internal shelf

The shelf creates a raised grow zone above the air-distribution and drainage area.

- Position the shelf about 3-5 in above the new bottom.
- Mark four to six mounting points.
- Use the smallest practical holes and large load-spreading washers.
- Seal all penetrations with silicone and allow full cure.
- Load-test with more weight than the expected substrate.
- Verify the tray slides in and out without removing the shelf.

6. Create passive fresh-air openings

Begin conservatively because holes can be enlarged later.

- Use approximately eight filtered openings.
- Place roughly two on each side near substrate or tray height.
- Space openings instead of clustering them.
- Avoid placing all openings at the top.
- Tune later by changing filter density or partial coverage.

7. Install the circulation fan

The fan should circulate air, not blast the grow surface.

- Mount high on the rear or side wall.
- Aim airflow along a wall rather than at the substrate.
- Use a chamber-side guard and vibration-isolating washers.
- Use a sealed cable gland for the wire.
- Set the fan to the lowest practical speed.

BUILD NOTE: At tray height, airflow should feel gentle rather than drafty.

8. Install the humidity inlet

Introduce fog indirectly so it disperses before reaching the grow tray.

- Place the inlet low on one side or below the shelf.
- Direct the inlet toward a wall or under-shelf plenum.
- Install the bulkhead with gasket and sealant.
- Route the hose without loops, dips or low spots.
- Whenever practical, slope the hose back toward the humidifier.
- Secure the hose so it cannot pull on the tub wall.

9. Place the humidity probe

The probe should read the chamber environment rather than the fog plume.

- Mount near tray height and toward the center.
- Keep several inches away from the fog inlet.

- Avoid direct fan airflow and dripping surfaces.
- Use a removable mount for cleaning and calibration.

10. Add basic drainage

Keep drainage simple for the prototype.

- Start with a removable drip tray under the shelf.
- Optionally add a low-point drain and external bottle later.
- Inspect corners, hose entry, lid gasket and shelf mounts after several hours.
- Do not run unattended until water accumulation points are understood.

Power architecture

Solar panel -> AC70 solar input -> AC70 battery/inverter

AC70 AC outlet -> watt meter -> power strip -> Inkbird -> humidifier

AC70 USB/DC output -> compatible fan and ESP32/sensors

11. Assemble the power system

- Keep the battery outside the humidified chamber.
- Elevate AC connections and create drip loops in every cord.
- Keep electrical components away from the drain bottle.
- Protect solar cables from doors, sharp edges and standing water.
- Do not connect a fan to 12V until its input specification is verified.

12. Run a 24-hour empty-chamber test

Operate the system without substrate before beginning a live grow.

- Record humidity rise time and overshoot.
- Locate condensation and hose water traps.
- Check whether one area receives excessive airflow.
- Watch for direct fogging of the probe.
- Record battery percentage after 24 hours.
- Record solar energy recovered during daylight.

BUILD NOTE: Begin with a wider humidity differential to avoid rapid on/off cycling.

13. Measure the real energy budget

Record both instantaneous power and full-day energy use.

- Fan watts
- Humidifier watts while active
- Controller and sensor watts
- Total idle watts
- Total watts during humidification

- Humidifier run time over 24 hours
- Overnight battery percentage used
- Solar watt-hours produced

DAILY ENERGY USE = AVERAGE SYSTEM WATTS x 24 HOURS

Example: 20 W average x 24 h = 480 Wh/day

14. Tune airflow and humidity

Change only one variable at a time and log the result.

Observed condition	Adjustment sequence
Chamber too wet	Lower target; widen differential; increase passive exhaust slightly; reduce humidifier output; improve drainage
Surface drying	Lower fan speed; aim fan at wall; add diffuser; reduce nearest opening; move tray
Top-to-bottom variation	Redirect fog inlet; improve under-shelf distribution; slightly increase circulation; reposition probe

15. Perform simulated failure tests

Confirm the prototype fails safely before introducing a live grow.

- Disconnect solar and confirm battery operation continues.
- Turn off humidification and measure humidity decay.
- Turn off the fan and confirm passive airflow remains available.
- Test probe disconnection behavior safely.
- Simulate battery shutdown and restore power.
- Confirm every required component restarts automatically.

16. Run a one-week biological test

Use one manageable grow tray and observe the system twice daily.

- Temperature and RH
- Battery state and solar input
- Humidifier duty cycle
- Condensation and surface moisture
- Drying patterns and growth uniformity
- Contamination signs
- Daily maintenance time

BUILD NOTE: Keep AI control disabled during Phase 1. First learn what normal operation looks like.

17. Decide whether Phase 1 passed

Freeze the Phase 2 cabinet specifications only after the prototype meets all success criteria.

Recommended build schedule

Timing	Work package
Day 1	Finalize parts and bench-test electrical equipment
Day 2	Rotate the tub and install the internal shelf
Day 3	Drill passive openings and install the fan
Day 4	Install humidity plumbing and probe
Day 5	Assemble battery and solar power
Day 6	Run the empty 24-hour test
Days 7-10	Tune airflow, humidity and drainage
Days 11-17	Run one live grow tray and collect data
Day 18	Review results and freeze Phase 2 specifications

PHASE 1 RULE: PROVE THE CLIMATE SYSTEM FIRST. MAKE IT BEAUTIFUL SECOND.